

GURU TEGH BAHADUR PUBLIC SCHOOL, DURGAPUR

SESSION : 2026-27

PRE-BOARD EXAMINATION – I

CLASS : XII

MAXIMUM MARKS : 50

SUBJECT : INFORMATICS PRACTICES (065)

TIME ALLOWED : 1 Hr. 45 Min.

General Instructions:

1. The paper is divided into 5 Sections - A, B, C, D and E.
2. All questions are compulsory.
3. All programming questions are to be answered using Python Language only. Assume that the pandas library has been imported as `pd` wherever required.
4. In case of MCQs, text of the correct answer should also be written.
5. Write all the sub-parts of a question together sequentially.

SECTION - A

Q.1 Answer the following: **(1 X 13 = 13)**

(a) State True or False:

"The index labels of a Pandas Series must always be integers."

(b) What will be the output of the following code?

```
s = pd.Series([10, 20, 30, 40, 50])
print(s[1:4].sum())
```

(A) 60

(B) 90

(C) 100

(D) 120

(c) Which attribute of a Series returns the total number of elements in it, including NaN values?

(A) count

(B) len

(C) size

(D) index

(d) Which of the following functions is used to read data from a CSV file into a DataFrame?

(A) `pd.read_csv()`

(B) `pd.to_csv()`

(C) `pd.load_csv()`

(D) `pd.open_csv()`

(e) Which parameter of `to_csv()` is used so that the row labels of a DataFrame are not written to the CSV file?

(A) `header=False`

(B) `index=False`

(C) `sep=None`

(D) `labels=False`

(f) In a DataFrame, `axis=1` refers to the _____ of the DataFrame.

(g) Riya wants to transfer pictures from her mobile phone to her laptop. She uses Bluetooth Technology to connect two devices. Which type of network will be formed in this case?

(A) PAN

(B) LAN

(C) MAN

(D) WAN

(h) Which of the following is the correct protocol for phone calls over the Internet?

(A) PPP

(B) FTP

(C) HTTP

(D) VoIP

(i) Which of the following networking devices is used to regenerate and transmit the weakened signal ahead?

- (A) Hub
- (B) Ethernet Card
- (C) Repeater
- (D) Modem

(j) Which of the following statements is false?

- (A) SMTP and POP protocols are used in email communication.
- (B) URL of a page is not always the same as its domain name.
- (C) HTTPS is safer than HTTP.
- (D) Interlinking of collection of webpages is called Internet.

(k) In which network topology is every node connected directly to every other node of the network?

- (A) Star
- (B) Bus
- (C) Tree
- (D) Mesh

Q1 (l) and (m) are Assertion (A) and Reason (R) based questions. Mark the correct choice as:

- (A) Both A and R are True and R is the correct explanation for A.
- (B) Both A and R are True and R is not the correct explanation for A.
- (C) A is True but R is False.
- (D) A is False but R is True.

(l) **Assertion (A):** When two Series having different index labels are added, the result contains NaN for the labels that do not match.

Reason (R): Arithmetic operations between two Series are performed on matching index labels.

(m) **Assertion (A):** A website can remember the language and theme chosen by a user on the next visit.

Reason (R): Cookies are small text files that a website stores on the user's computer through the web browser.

SECTION - B

Q.2 Complete the given Python code to get the required output as: **Rajasthan** (2)

```
import _____ as pd
di = {'Corbett': 'Uttarakhand', 'Sariska': 'Rajasthan',
      'Kanha': 'Madhya Pradesh', 'Gir': 'Gujarat'}
NP = _____.Series( _____ )
print(NP[ _____ ])
```

Q.3 Differentiate between Web server and Web browser. Write any two popular web browsers. (2)

Q.4 Carefully observe the following code: (2)

```
import pandas as pd
Year1 = {'Q1': 5000, 'Q2': 8000, 'Q3': 12000, 'Q4': 18000}
Year2 = {'A': 13000, 'B': 14000, 'C': 12000}
totSales = {1: Year1, 2: Year2}
df = pd.DataFrame(totSales)
print(df)
```

Answer the following:

- (i) List the index of the DataFrame df.
- (ii) List the column names of the DataFrame df.

SECTION - C

Q.5 The four houses of Class 12-D of a school are Beas, Chenab, Ravi and Satluj, having 18, 2, 20 and 18 students respectively. Write Python statements to: (3)

(a) Create a Series object **House** from a dictionary that stores the number of students in each house. (2)

(b) Display the names and strength of only those houses that have more than 15 students. (1)

Q.6 Differentiate between the following (one point each): (3)

(a) Website and Web page

(b) Static web page and Dynamic web page

(c) Hub and Switch

Q.7 Consider the following DataFrame **df**: (3)

	Name	Eng	Maths
S1	Asha	78	88
S2	Bimal	65	72
S3	Chitra	90	95
S4	Dev	55	40

Write the output of the following:

(a) `print(df['Eng'] + df['Maths'])`

(b) `for i, row in df.iterrows():`
 `if row['Eng'] > row['Maths']:`
 `print(i, row['Name'])`

(c) `print(df.Maths.max(), df.Eng.min())`

SECTION - D

Q.8 Consider the following list: (4)

```
[[ 'Divya', 'HR', 95000], [ 'Mamta', 'Marketing', 97000],  
[ 'Payal', 'IT', 98000], [ 'Deepak', 'Sales', 79000]]
```

Write Python statements to:

(a) Create a DataFrame **Emp** from the given list with the column headings Name, Department and Salary. (2)

(b) Export the DataFrame **Emp** to a CSV file named "EMP.csv" without writing its row labels. (1)

(c) Read the file "EMP.csv" back into a new DataFrame named **Emp2**. (1)

Q.9 Answer the following questions: (4)

(a) Expand the following: URL, WWW

(b) What is web hosting?

(c) What are add-ons (plug-ins) in a web browser? Give one example.

(d) Name the device that converts digital signals into analog signals and vice versa. Why is it needed?

Q.10 Consider the given DataFrame **Genre**:

(4)

	Type	Code
0	Fiction	F
1	Non Fiction	NF
2	Drama	D
3	Poetry	P

Write suitable Python statements for the following:

- (a) Add a column called **Num_Copies** with the values 300, 290, 450, 760.
- (b) Add a new genre of type 'Folk Tale' having code 'FT' and 600 number of copies, with row label 4.
- (c) Delete the row of the genre 'Drama' (row label 2).
- (d) Change the Num_Copies of 'Poetry' (row label 3) to 500.

SECTION - E

Q.11 Ekam, a Data Analyst with a multinational brand, has designed the DataFrame **df** that contains the four quarters' sales data of different stores as shown below:

(5)

	Store	Qtr1	Qtr2	Qtr3	Qtr4
0	Store1	300	240	450	230
1	Store2	350	340	403	210
2	Store3	250	180	145	160

Answer the following:

- (a) Predict the output of: `print(df.size)`
- (b) Write a Python statement to add a new column **Total Sales**, which is the sum of all four quarters' sales.
- (c) Write a Python statement to display the store-wise highest quarterly sale (row-wise maximum of Qtr1 to Qtr4).
- (d) Write a Python statement to delete the last row from the DataFrame.
- (e) Write a Python statement to export the DataFrame to a CSV file named "data.csv" stored in the D: drive.

- Q.12** 'Swabhaav' is a big NGO working in the field of Psychological Treatment and Counselling, (5) having its Head Office in Nagpur. It is planning to set up a center in Vijayawada. The Vijayawada Center will have four blocks – ADMIN, PSYCHIATRY, PSYCHOLOGY and ICU. You, as a Network Expert, need to suggest the best network-related solutions for them to resolve the issues/problems mentioned in questions (a) to (e), keeping the following parameters in mind:

From	To	Distance
ADMIN	PSYCHIATRY	65 m
ADMIN	PSYCHOLOGY	65 m
ADMIN	ICU	65 m
PSYCHIATRY	PSYCHOLOGY	100 m
PSYCHIATRY	ICU	50 m
PSYCHOLOGY	ICU	50 m

Block	No. of Computers
ADMIN	16
PSYCHIATRY	40
PSYCHOLOGY	19
ICU	20

Distance of Nagpur Head Office from Vijayawada Center = 700 km

- (a) Suggest the most appropriate location of the server inside the Vijayawada Center. Justify your choice.
- (b) Which hardware device will you suggest to connect all the computers within each block of the Vijayawada Center?
- (c) Draw a cable layout to efficiently connect various blocks within the Vijayawada Center.
- (d) Where should the router be placed to provide internet to all the computers in the Vijayawada Center?
- (e) Which type of network (PAN, LAN, MAN or WAN) will be set up among the computers connected with the Vijayawada Center?

***** End of Question Paper *****